

Institute of Actuaries of India

Subject SA7-Investment and Finance

November 2020 Examination

INDICATIVE SOLUTION

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i)

Lifestage Strategy

- The biggest problem with such strategy is to determining the attributes of an “average” member who may not really exist and trying to fit strategy to all the members without considering their individual needs like vacations/ home purchase/ childrens education and marriage/ medical needs etc.
 - Very few members will actually retire and may access their funds as cash – offer a cash portfolio as an alternative.
- Certain members may be more or less risk tolerant depending on their particular investment horizon and personal circumstances.- opportunity may be given to “opt out” of the Lifestage structure and phasing down process described and either select one of the risk profile portfolios offered or alternative portfolios could be offered.
- strategy members will take some part of their corpus as cash lumpsum. Tax incentives may reinforce this behaviour. Re-design a strategy which does offer a pre-retirement portfolio targeting full annuitisation of capital at retirement. Members could then “opt-out” their cash lumpsum.
- Focus on preservation of capital and not preservation of income as members approach retirement.
 - Protect the size of annuity that can be purchased rather than the capital amount at retirement.
 - Consider the types of annuities members typically purchase at retirement.
 - A fixed annuity would require a matching portfolio of bonds of the correct duration
 - With profit annuity or living annuity would require a more aggressive approach
 - introduce different pre-retirement channels catering more specifically to popular annuity choices.
 - administratively complex but also forces members to consider their situation and options in good time,
- Too conservative at retirement and de-risking starts too early and is too severe considering the potential longevity of members who are retiring at 60– potentially a 30 year time horizon for younger member.
- Pre-retirement portfolio may be considered too volatile by some and members may in fact lose money as they approach retirement.
- This strategy with fixed asset class allocation does not take current market conditions into account and doesn’t allow managers discretion in asset classes and members thus loses on extra returns that could have been achieved by actively managing the funds.
- Funds for a members are mechanically switched and could be selling and buying at worst possible time when not necessary. This is especially true where we have volatile asset classes.
- The solution would be to introduce a more gradual phasing in from one portfolio to the next, Trying to time the markets and make a call on the timing of the moves is not a recommended solution.
 - For example mechanically switching from a higher risk portfolio to a lower risk portfolio (eg High to Medium risk or Medium to Low risk) could take place on a specific date or during period where equity markets have experienced a significant correction.
 - These losses are then crystallized as a result of the switch and the member misses out on the subsequent recovery on this portion of his assets if the switch had been delayed – often by a relatively short period.
- The administration and management framework required to run a more complex strategy such as this one,
 - Together with more frequent trading associated with the switches between the Lifestage portfolios could result in increased direct costs being borne by members.

[10]

ii) **Unquoted Securities**

Advantages:

- Diversification, although if the unquoted companies operate in the similar industries to quoted companies in which the fund invests there may be less diversification than first thought.
- Potentially higher returns as the security may be less well researched, leading to pricing anomalies.
- Unquoted companies are often small and therefore can still grow rapidly even in relatively mature industries.

Disadvantages:

- The risks will be higher as often the company and the management will have no track record.
- The investment will be very illiquid and may need to be held for long time.
- It will be difficult to value.
- The company may be more dependent on one person.
- It will require specialist knowledge to invest in this area, this will mean higher management fees. It is unlikely that the in-house team will possess the necessary skills, therefore a third party fund manager will need to be employed, thus increasing costs.
- The companies are likely to be less financially stable and will have less ability to raise additional capital should the need arise.

Commodities

Advantages:

- Investment in commodities would provide diversification for the pension fund.
- It is possible to make significant profits from investing in commodities in short periods, however there is also the opportunity to lose large amounts in equally short periods.

Disadvantages:

- Like investing in unquoted securities, the Trustees will almost certainly need to employ third party fund managers.
- Use of commodity shares (mining, exploration companies) gives less diversification from equity market than physical would.
- Institutions do not invest directly in commodities as this would involve shipping and storage of large amounts of material and institutions do not have the necessary skills or facilities.
- Commodities do not naturally fit into an asset liability model as they are neither real assets or fixed rate assets, therefore unless the Trustees can identify an institution with a very good record in this area there is little to justify investing in this area.

[7]

iii) **a) Domestic Equities**

With lockdowns, lot of manufacturing, retail and auto companies would have found it hard to continue without any business interruptions. This would will hurt the performance of domestic equities. If Govt announces any measures / initiatives to simulate growth there may be offset to some degree. Negative sentiments may affect the equity market

[2]

Interest rates are unlikely to be lowered because of inflation, and would not act in their usual counter-cyclical way to help equities during times of poor economic growth.

[1]

The domestic economy is likely see its currency depreciate. However, higher interest rates (to combat inflation) may stimulate the flow of money into the economy which help strengthen the currency.

A depreciating currency will see domestic companies with operations abroad, see the value of the operating profits of those operations rise in local currency terms and this should benefit their share prices.

[2]

Share prices will also be helped by the rising real value of their assets.

[Max 4]

b) Commodities

In theory, the price of commodities like Aluminum should rise in line with general price rises. It should maintain its real value.

[1]

The demand for Aluminum for industrial, and other economic activity related uses, is likely to fall due to the economic stagnation.

[1]

This will likely negatively affect the price of Aluminum and to a greater degree than for gold as Aluminum is used to a greater extent in industry and manufacturing.

Commodities may benefit from Government related economic stimulus.

[2]

However, government sponsored grants and subsidies (if any) may be scaled back due to likely fiscal problems.

[1]

[Max 3]

iv)

Annuitisation Options

a) Fixed Life Annuity (with or without fixed increases)

- Member purchases an annuity from insurer. Annuity is guaranteed to be paid for life of annuitant and may be designed to pay a percentage income to a qualifying widow/er on death of the annuitant. Annuity amount remains fixed throughout the life of the annuitant, reducing for the spouse. No annual increases are awarded.
- On death of spouse no further payment is made
- No longevity risk but significant inflation risk
- Costs associated with writing guaranteed annuity business are high for insurers – expensive option

[2]

b) Living Annuity

- Member invests their Fund credit with a life insurer. Member then draws down a monthly income from their investment at a level of their choice. Regulation may limits this to between say 1% and 20% of the capital balance every year.
- Member has a degree of choice around the investment strategy and portfolios for their fund value.
- Any remaining balance in the account at death is paid to estate/beneficiaries
- Member assumes longevity risk – if funds run out before death he will run out of money.
- Member assumes investment risk – if asset perform poorly it may affect his level of income

[2]

c) Inflation-linked annuity

- As for level annuity, but annuity amount is guaranteed to increase with inflation every year
- Therefore removes inflation risk as well as longevity, investment risk
- Due to high level of guarantees offered by the Life Company/issuer, this is the most expensive type of annuity to secure

[2]

d) With Profits annuity

- As for level annuity, but annuity amount will increase every year by an amount declared by the issuer. This will depend on the performance of the underlying assets as well as a degree of smoothing.
- Increases are therefore not guaranteed
- Amount of annuity cannot however decrease from one year to the next.
- A popular alternative to a inflation linked annuity as the costs may be lower due to the lower degree of guarantee offered

Member may still assume some inflation risk – particularly when market conditions are poor

[1]

v)

Inflation linked annuity (ILA)

- Pricing of ILA closely linked to (longer dated) inflation linked bonds (ILBs).
- In order to protect the level of income that can be purchased, fund should have a significant allocation to a portfolio of ILBs
- Consider appropriate duration of the ILB portfolio
- This represent the duration and yield curve sensitivity of the insurer's ILA underlying portfolio.
- Hence would need to take care in how to measure the performance of the ILB portfolio as it may not be represented by a typical index.
- Exposure to ILB's will also protect the (real) value of capital if ILB yields are not too volatile – may be more appropriate for the portion intended to be taken in cash at retirement
- Yields on ILB's are low, so may want some exposure to growth assets (such as equities) given the period of time in this portfolio
- With Profit annuities – these policies usually back by a range of assets such as equities, bonds, ILB's.
- Risks are pooled and returns are effectively "smoothed" in order to provide a sustainable predictable level of pension increase
- Pricing of with profit annuities not closely linked to a particular asset class. Prices usually more static (given the pooling/smoothing)
- A combination of cash, equities and bonds – not unlike what fund has right now – would be appropriate
- Will still need to protect the capital for a portion of the assets intended to be taken as cash at retirement

[5]

vi)

Volatility Index**a) VIX is index based on implied volatilities of options and it is forward looking**

- i.e. it incorporates investors' assumptions about future volatility
- Volatility is a measure of risk in financial markets. It estimates how far prices are expected to move in a given time frame. The VIX is an expectation or forecast of the markets perception of risk in capital market in India

VXI provides investors with a measure of equity market sentiment on the relevant stock indices (NIFTY, SENSEX)

- When volatility is low, there is a measure of complacency evident in the markets; lack of fear.
- High volatility, however, suggests a fearful market.

[3]

b) VIX enables investors to monitor this "sentiment" forecast on a daily basis.

VIX can also be used as a country/political risk measure

- The difference between the VIX and US S&P 500 VIX approximates the equity market “fear premium” between an emerging market and a first world equivalent
 - High risk differentials are associated with weak INR/USD

VIX is useful as a timing tool or directional indicator

- Spikes in the VIX are accompanied by troughs in the NIFTY50 (signalling a buying opportunity)
- However, although the NIFTY50 typically rallies after sharp upward moves in implied volatility, low levels of the VIX do not imply a weaker NIFTY50 market

The VIX Future enables volatility to be traded as a separate asset class

- VIX futures are negatively correlated with the NIFTY
- Consequently they can be used as a hedge against equity market falls
- Studies show the benefit of combining volatility and equities to manage downside risk

Volatility futures can provide an alternative hedging instrument to put options based on their relative value (for a given implied volatility term structure)

[3]

[42 Marks]

Solution 2:

i)

The behaviors exhibited are

a)

- Anchor and adjustment – Has view about stock and continues to hold view even where news outlook is negative.
- Confirmation bias – look for evidence to confirm their views even though the market says something different
- Regret aversion – by retaining the existing arrangement will not suffer from regret if they sell share and the value actually goes up rather than down
- Status Quo bias – people have preference to keep things as they are

[2]

b)

- Confirmation bias – looks for evidence that confirms their point of view
- Anchor and adjustment – has pre-defined idea of stock, it is favourite company, so doesn't matter what people say they will continue to believe it will do well.

[2]

c)

- Anchor and adjustment – has view of stock and uses past performance to make judgement. View on growth is anchored on past performance
- Overconfidence – investor overconfident in their ability to predict future cashflows
- Availability – easier to imagine future returns based on past experience

[2]

d)

- Framing – although the actual expected return of the alternative bond is higher the investor has framed the decision in such a way as to make the alternative bond seem poor value even though it offers the better return.
- Regret aversion – if hold the alternative investment and the return actual is 0% will regret not holding the guarantee return.
- Risk adverse/myopic loss aversion – prefers to hold a guarantee return as places more weight on the downside return of the alternative investment rather than the upside
- Representative Heuristics – by having a guarantee return they place a higher value than an alternative that involves chance. By adding further probabilities into the scenario the return profile becomes more difficult to understand

[2]

ii)

a) Two example of prescriptive regulations

- The investment in Mutual Funds AT ANY POINT OF TIME shall not exceed 50%* of Investment falling under "Other than Approved Investments" for both Life and General Insurance Companies
- Companies whose preference shares are selected for investment; dividend should have been paid on equity shares for two years out of immediately preceding three years.

[Any other regulations is also ok]

[2]

b)

Advantages

- They are open to less abuse
- They give more public confidence

Disadvantage

- Cost of regulatory compliance becomes high
- there could be a trade off in the returns that could be achieved due to such regulations

[4]

c) A minimum reserve requirement is a requirement to hold a certain percentage of deposits in liquid assets (e.g. cash or treasury bills). This is necessary to ensure that a bank that accepts deposits will be able to meet demands for withdrawals from depositors. This measure is generally considered necessary to provide depositors with confidence in the banking system.

A risk-based capital requirement differs from the above by requiring a bank to have capital support to cover a risk-based measure of its assets (loans and deposits with other institutions). This is different to the solvency requirement of ensuring assets exceed liabilities, which would also need to be met.

[3]

d) A non risk-based capital measure such as solvency would be successful in ensuring that assets exceed liabilities, however this ignores the complexities of valuing a portfolio of loans.

Loans will either be paid back at the level set out in the documentation, or they will default.

In the majority of scenarios, a loan would be paid in full. However, in a significant number of scenarios, the proceeds from the loan (which is an asset of the bank) will be much less than this.

If there is no penalty for risk in the capital measure, then a bank will have an incentive to write a larger proportion of lower quality (but higher expected profit) loans within its loan book to maximise its expected profits for a given amount of capital deployed.

As banks are a key part of financial system, involved in both payments and deposit-taking, it is generally accepted that a risk-based capital measure is preferable.

[5]

[22 Marks]

Solution 3:

i)

The acquisition might be a good use of cash because

- Lead Life is taking greater control of the distribution of its products to market.

- Keeping a significant cash balance may make Lead Life a takeover target itself
- Lead Life could return its cash to its stockholders through a dividend payment.
- Therefore, the acquisition is sensible if Lead Life can obtain a better return with this investment
- Merging the companies may bring efficiencies through co-ordination and administration.
- Areas where efficiencies may be sought are in:
IT , Finance and Accounting, Legal services and Compliance , Marketing, where the merger is likely seek to reinforce a single brand and therefore will want to achieve consistency..
- Lead Life may consider that the management of Aam Bazaar is poor and believes it can manage the business more effectively given its knowledge of the target market.
- However, it may not be possible to replace the management wholesale depending on the governance in place at Aam Bazaar.
- In reality, Lead Life management may be inexperienced in dealing with the issues Aam Bazaar has faced and be unable to correct the perceived failings.
- Lead Life might be seeking to diversify away from being a principal to being a service provider as well.
- However, stockholders may not be attracted to this aspect as they can achieve diversification more easily through their own investment strategy.
- With larger scale, Lead Life may be able to achieve lower financing costs.
This would be achieved by offering larger volumes of securities. at a lower interest rate due to the additional size of the company.
- However, lower financing costs are achieved because the two entities are now guaranteeing each other's debts.
- This leads to a lower risk premium in the valuation of debt securities.
- However, stockholders option to default has reduced in value.
- Issued bonds of Aam Bazaar will gain in value due to the lower perceived risk.
- This has been paid for by the stockholders in the new combined entity.

[1 mark for each – max 12]

[12]

ii) The various strategies that can be adopted by Aam Bazaar are:-

- The Board of Aam Bazaar could change its charter so that only a portion of the Board is able to be elected each year.
- Aam Bazaar could apply restricted voting rights to shareholders that own more than a specified percentage of the stock.
- Aam Bazaar could decide to buy back the shares purchased by Lead Life at a higher price than Lead Life paid under the stipulation that Lead Life does not pursue the bid.
- Aam Bazaar could require that a waiting period is enforced before any acquisition can complete.
- The debt of Aam Bazaar could be structured to contain a clause that requires immediate repayment of the debt if there is a hostile change of control.
- Existing shareholders can be issued rights that allow them to buy additional shares at a low price if there is a significant purchase of shares by a hostile bidder.
- Aam Bazaar could acquire shares in Lead Life itself. This could allow Aam Bazaar to try and block the transaction Lead Life is pursuing.
- It may also acquire assets that Lead Life doesn't want or would be problematic, for example if caused asset admissibility problems for solvency purposes.
- Aam Bazaar could look for a different purchaser or partner that it finds more attractive. Such an entity is often referred to as a White Knight. The merged entity would have an increased market capitalisation that would make it more expensive for Lead Life to acquire.
- Lead Life may not want the business that the White Knight brought which would make it less attractive.

[6]

- iii) Share prices will be affected by:
- Market movements
 - Markets evaluation of the deal – both for the return it will generate for Lead life
 - Also the reading of the market w.r.t the price Lead Life is planning to pay
 - Size of the transaction for Lead Life and thereby impact on its balance sheet
 - Dividends it may loose out to pay its S/H
 - Whether market sees the fit for the transaction
 - If the transaction proves to be lengthy and taking time for regulatory approvals that leads to value being destroyed
 - Any tax complications
 - Any arbitrage opportunities

[4]

- iv)
- Lead Life will need to construct a security whose income depends on the regular commission streams it is receiving.
 - The key factor in the level of commission that it receives on a block of business will be the persistency of that business and the level of renewals.
 - The company could structure the security as a multi-class security.
 - The block of business to be securitized could be arranged by business type, for example Lead Life and general business separately, where different persistency levels are experienced.
 - The most persistent business would be allocated to a tranche that was paid first.
 - The number of classes would be dependent on the number of sensible homogenous groups that could be constructed.

[Max 4]

v) And vi)

Cut back its activities	Can lead to losing out customers
Look for alternative sources of capital like strategic investment	With the economy getting impacted due to COVID, the alternate source investments take time to fructify
Evaluate each of the projects financed and choose which is most relevant to continue	Incremental Returns generated may be compromised
Seek for injection of capital	Shareholders / Stockholders may not be keen
Take a higher investment risk to generate some short term return	regulatory risk
Look for another potential buyer	Market reaction may not be positive and thereby price may get impacted
Focus on core business	Loose out to competition and perhaps
Advertising and marketing	Loose out the position in the market which may affect sales

[Max 5 + 5=10]

[36 Marks]
